Maharashtra HSC Exam, 2016
Chemistry

Time-3 Hours **M.M- 70**

General Instructions:
1. All questions are compulsory.
2. Answers to the questions of Section-I and Section –II should be written in the same answer book.
3. Draw neat, labelled diagrams and write balanced chemical equations wherever necessary.
4. Figures to the right indicate full marks.
5. Use of logarithmic table is allowed.
6. Every new question must be started on a new page.

Section-1

Q.1. Answer any SIX of the following: [12]

i. What is ferromagnetism? Iron (Z = 26) is strongly ferromagnetic. Explain.

ii. Define boiling point. Write the formula to determine molar mass of a solute using freezing point depression method. iii. Write mathematical equations of first law of thermodynamics for the following processes: a. Adiabatic process
 b. Isochoric process

iv. Explain graphical method to determine activation energy of a reaction.

v. Write the names and chemical formulae of any one ore of iron and zinc each.

vi. What is the action of
 a. Sodium on arsenic?
 b. Magnesium on bismuth?

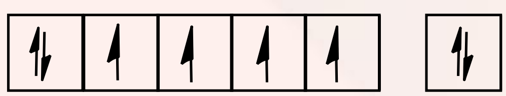
vii. Define enthalpy of sublimation. How is it related to enthalpy of fusion and enthalpy of vaporization?

viii. What are Ellingham diagrams? Write any two features of it.

Solution 1:

(i) The substance containing large number of electrons can be permanently magnetised, this property is known as ferromagnetism.

${}_{26}Fe = [AR]3d_6 4s_2$



Iron consist of four unpaired electrons, hence it is strongly ferromagnetic.

(ii) The temperature at which vapour pressure of liquid is equal to atmospheric pressure is called as boiling point.

$$\Delta T_f = K_f \times \frac{W_2 \times 1000}{W_1 \times W_2}$$

(iii) (a) In adiabatic process, there is no exchange of heat thus, $q = 0$. So, First law becomes $\Delta U = w$.

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(b) Isochoric process, as it is carried in closed vessel $\Delta V = 0$ thus $q = 0$. So, First law becomes $\Delta U = q$.

(iv) Arrhenius equation in

$$\ln k = \ln A - \frac{Ea}{RT}$$

Taking log 10, $\log_{10} k = -\frac{Ea}{RT} + \log_{10} A$

$\log_{10} k$ is plotted against $1/T$, the graph is straight line.

In graph slope $= A - \frac{Ea}{2.303R}$

$$\Rightarrow Ea = -\text{slope} \times 2.303 \times R$$

(v) (a) **Sodium on arsenic:** Arsenic reacts with sodium metal to form sodium arsenic.

$2As + 3Na \rightarrow Na_2As_3$ (Not expected)

(b) **Magnesium on bismuth:** Bismuth reacts with magnesium to form magnesium.

$2Bi + 3Mg \rightarrow Mg_3Bi_2$ (Not expected)

(vii) The enthalpy change that accompanies is called as enthalpy of sublimation.

$\Delta_{sub}H = \Delta_{fus}H + \Delta_{vap}H$

(viii) The graph plotted ΔG^{o} against T(K) for the formation of oxide is called as Ellingham Diagram.

Features:

1) The graph for the formation of metal oxide is a straight line with an upward slope.
2) There is sudden change in the slope of some metal oxides such as MgO, ZnO and HgO.
3) For a few metal oxide of mercury and silver the graph is at the upper part in the Ellingham diagram (AgO and HgO).

Q.2. Answer any THREE of the following: [9]

(i) Silver crystallises in fcc structure. If density of silver is 10.51 g cm^{-3}, calculate the volume of unit cell. [Atomic mass of silver (Ag) = 108 g mol^{-1}]

ii. The vapour pressure of pure benzene is 640 mm of Hg. 2.175×10^{-3} kg of non-volatile solute is added to 39 g of benzene, the vapour pressure of solution is 600 mm of Hg. Calculate molar mass of solute (C = 12, H = 1).

(iii) Calculate C−Cl bond enthalpy from the following reaction:

$CH_3Cl_{(g)} + Cl_{2\,(g)} \rightarrow CH_2Cl_{2(g)} + HCl_{(g)}$; $\Delta H° = -104$ kJ

If C−H, Cl−Cl and H−Cl bond enthalpies are 414, 243 and 431 kJ mol^{-1} respectively.

(iv) Define cell constant. Draw a neat and well labelled diagram of primary reference electrode.

Solution 2:

(i) Mass of 1 atom of Ag

$$= \frac{\text{Atomic mass of Ag}}{N_A}$$

$$= \frac{108}{6.033 \times 10^{23}}$$

$$= 17.03 \times 10^{-23}$$

Mass of unit cell

$= \text{Atomic in unity cell} \times \text{Mass of 1 atom}$

$= 4 \times 17.03 \times 10^{-23}$

$= 71.71 \times 10^{-23}$

Volume of unit cell

$$= \frac{\text{Mass of unit cell}}{\text{Density}}$$

$$= \frac{71.71 \times 10^{-23}}{10.51}$$

$= 68.24 \times 10^{-23}$

(ii) Molar mass of benzene

$= (6 \times 12 + 6 \times 1)$

$= 78 \times 10^{-3}\,\text{kgmol}^{-1}$

$P^o = 640\text{mm}, P = 600\text{mm Hg}$

$W_1 = 39 \times 10^{-3}\text{kg}, W_2 = 2.175 \times 10^{-3}\text{kg}$

$M_1 = 78 \times 10^{-3}\text{kg}, M_2 = ?$

$$M_2 = \frac{P^o \times W_2 \times M_1}{\Delta P \times W_1}$$

$$M_2 = \frac{640 \times 2.175 \times 10^{-3} \times 78 \times 10^{-3}}{(640 - 600) \times 39 \times 10^{-3}}$$

$= 69.6 \times 10^{-3}\,\text{kgmol}^{-1}$

(iii)

$\Delta H^o = \sum H^o_{\text{reactant bond}} - \sum H^o_{\text{product bond}}$

$\Rightarrow -104 = (3 \times H^o_{C-H} + H^o_{C-Cl} + H^o_{Cl-Cl}) - (2 \times H^o_{C-H} + H^o_{C-Cl} + H^o_{Cl-Cl})$

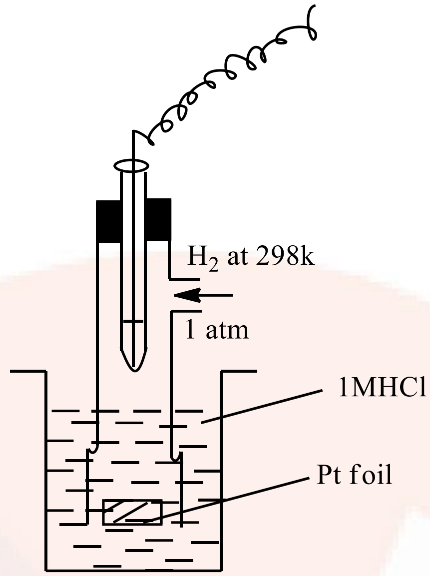
$\Rightarrow -104 = (3 \times 414 + X + 243) - (3 \times 414 + 2X + 431)$

$\Rightarrow -104 = (1242 + X + 243) - (828 + 2X + 431)$

$\Rightarrow -104 = 226 - X$

Thus, X = 330kJ mol^{-1}.

(iii) **Cell constant:** The ratio of the distance between the electrodes divided by the area of cross-section of the electrode is called as cell contant.

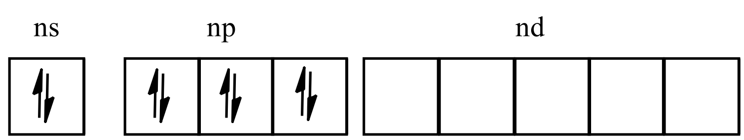


Standard or Normal Hydrogen Electrodes

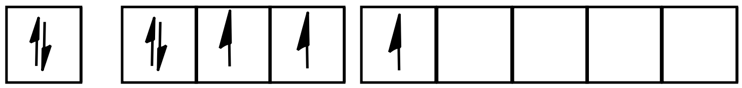
Q.3. Answer any ONE of the following: [7]
(A) Write four points of differences between properties of nitrogen and other elements of group 15.
Explain the structure of ClF_3.
Conductivity of a solution is $6.23 \times 10^{-5}\ \Omega^{-1}cm^{-1}$ and its resistance is 13710 Ω. If the electrodes are 0.7 cm apart, calculate the cross-sectional area of the electrode.
Why is molality of a solution independent of the temperature?

(B) What are neutral oxides? Explain the nature of zinc oxide with the help of the reactions.
Define 'molar conductivity' and 'zero order reaction'.
In a first order reaction x → y, 40% of the given sample of compound remains unreacted in 45 minutes. Calculate rate constant of the reaction.
Solution 3:
(A) (1) At ordinary temperature nitrogen is gas while other elements are solids.
(2) Nitrogen exists as diatomic molecule but other elements of the group are tetra atomic.
(3) Nitrogen has tendency to form pπ- pπ bonds but other elements form pπ- dπ instead of pπ-pπ.
(4) Due to small size and high electro negatively nitrogen has tendency to form hydrogen bonds in hydride compounds but other compounds are not.
(5) Nitrogen shows wide range of oxidation state form -3 to +5 but other elements not.
Explain structure of CIF_3.

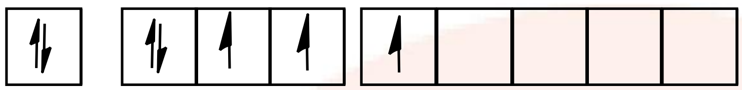
Ground state of $_{17}Cl$



Excited state of $_{17}Cl$

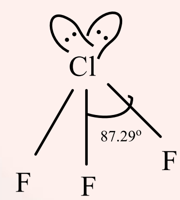


Hybridized State



sp^3d hybridization

Trigonal bipyramidal or T- shaped.



Cross-sectional area of electrode:

$$K = \frac{1}{R} \times \frac{1}{a}$$

$$\Rightarrow 6.23 \times 10^{-5} = \frac{1}{13710} \times \frac{0.7}{a}$$

$$\Rightarrow a = 0.8196 cm^3$$

(B) The oxides which are neither acidic nor basic are called as neutral oxides. Ex. NO, CO, N_2O, etc**.**

Amphoteric nature of ZnO

$$ZnO(s) + 2HCl(aq) \rightarrow ZnCl_2(s) + H_2 0(l)$$

$$ZnO(s) + 2NaOH(aq) \rightarrow NaZnO_2(s) + H_2 0(l)$$

Molar conductivity:

The conductance of a volume of a solution containing 1 mole of dissolved electrolyte when placed between two parallel electrodes 1 cm apart.

Zero order reaction:

The reaction whose rate is independent of the reaction concentration and remains constant throughout the course of the reaction.

Solution:

At t=45min,

Initial concentration $[A]_o$ = 100, final concentration $[A]_t$ = 40

$$K = \frac{2.303}{t}\log_{10}\frac{[A]_o}{[A]_t}$$
$$= \frac{2.303}{t}\log_{10}\frac{100}{40}$$
$$= \frac{2.303}{t}\log_{10} 2.5$$
$$= \frac{2.303 \times 0.3979}{45}$$
$$= 0.02036\,\text{min}^{-1}$$

Q.4. Select and write the most appropriate answer from the given alternatives for each Sub-question: [7]

i. The molecular formula $H_2S_2O_2$ represents which oxoacid among the following?

(a) Hydrosulphurous acid (b) Thiosulphurous acid
(c) Sulphuric acid (d) Pyrosulphurous acid

(ii) Iodine exists as

(a) Polar molecular solid (b) Ionic solid
(c) Non-polar molecular solid (d) Hydrogen bonded molecular solid

(iii) Absolute entropies of solids, liquids and gases can be determined by _______.

(A) measuring heat capacity of substance a t various temperatures
(B) subtracting standard entropy of reactants from products
(C) measuring vibrational motion of molecules
(D) using formula $\Delta S° = S^{\circ}_{T} - S^{\circ}_{0}$

(iv) The determination of molar mass from elevation in boiling point is called as _______.

(A) Cryoscopy (B) colorimetry
(C) Ebullioscopy (D) spectroscopy

(v) The process of leaching alumina, using sodium carbonate is called _______.

(A) Baeyer’s process (B) decomposition
(C) cyanide process (D) Hall’s process

(vi) On calculating the strength of current in amperes if a charge of 840 C (coulomb) passes through an electrolyte in 7 minutes, it will be _______.

(A) 1 (B) 2
(C) 3 (D) 4

(vii) A → B is a first order reaction with rate 6.6×10^{-5} M s^{-1}. When [A] is 0.6 M, rate constant of the reaction is _______.

(A) $1.1 \times 10^{-5}\ s^{-1}$ (B) $1.1 \times 10^{-4}\ s^{-1}$
(C) $9 \times 10^{-5}\ s^{-1}$ (D) $9 \times 10^{-4}\ s^{-1}$

Solution 4:

(i) (b) Thiosulphurous acid
(ii) (c) non-polar molecular solid
(iii) (a) measuring heat capacity of substance a t various temperatures.
(iv) (c) Ebullioscopy
(v) (d) Hall’s process

(vi) (b) 2

(vii) (a) $1.1 \times 10^{-5}\ s^{-1}$

SECTION – II

Q.5. Answer any SIX of the following: [12]

(i) Why is Sc^{3+} colourless while Ti^{3+} coloured? (Atomic number Sc = 21, Ti = 22)

(ii) Illustrate with example, the difference between a double salt and a coordination compound.

(iii) How is chlorobenzene prepared from aniline? How is chlorobenzene converted into diphenyl?

(iv) What is metamerism? Explain metamerism with suitable examples of ethers.

(v) What are ketones? How are ketones classified?

(vi) How are a. 1-nitropropane and b. 2- nitropropane prepared from suitable oxime?

(vii) Define antioxidants. Draw structure of BHT.

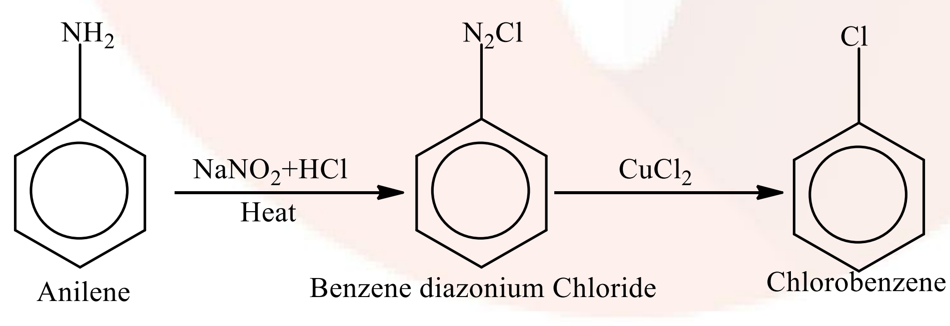
(viii) What are carbohydrates? Write the reaction for the preparation of nylon-6.

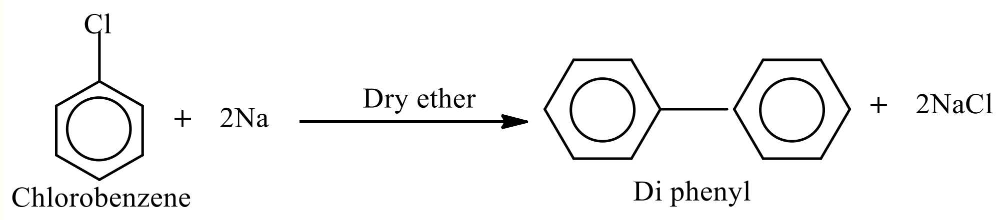
Solution 5:

(i) Electronic configuration of $_{21}Sc^{3+}$ is [Ar] $3d^0$ and $_{22}T^{3+}$ is [Ar]$3d^i$.
Therefore, due to absence of unpaired electron Sc^{3+} is colourless and due to presence of one unpaired electron d-d transition takes place hence Ti^{3+} is coloured.

(ii) Double salts are those molecular or addition compounds which exists in a solid state but dissociate into constituents ions when dissolved in water, Example, Mohr salt $FeSO_4(NH_4)_2SO_4.6H_2O$.
Coordination compounds are those molecular or addition compounds which retain their identity in aqueous solution and show property entirely different from their constituent ions. Example, Potassium ferrocyanide $K_4[Fe(CN)_6]$.

(iii)





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(iv) Metaners are either chain isomer or position isomers with same functional group and have different alkyl groups attached to heteroatom.
Following metamers are represented by formula $C_4H_{10}O$.

1. $CH_3CH_2OCH_2CH_3$ – di ethyl ether.
2. $CH_3OCH_2CH_2CH_3$ – methyl n-propyl ether.
3. $CH_3O\underset{\underset{CH_3}{|}}{C}HCH_3$ – methyl isopropyl ether

(v) First oxidation product of secondary alcohol or the organic compound containing keto or >C=O group in which both valency of carbonyl carbon are satisfied by alkyl or aryl radical.
Simple ketones, example, Acetone
Mixed ketone, example, Ethyl methyl ketone.

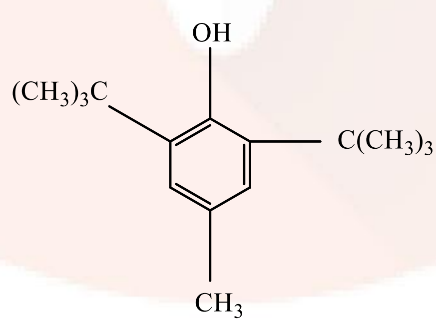
(vi) (a) 1- Nitropropane:

$$\underset{\text{Propyldoxime}}{CH_3CH_2-\overset{\overset{H}{|}}{C}=N-OH} \xrightarrow{\text{Triflouroacetic acid}} CH_3-CH_2-\overset{\overset{H}{|}}{C}=\overset{\overset{O^-}{|}}{N}-OH \rightleftharpoons \underset{\text{1-nitropropane}}{CH_3CH_2-\underset{\underset{H}{|}}{\overset{\overset{H}{|}}{C}}-\overset{\overset{O^-}{|}}{N}=O}$$

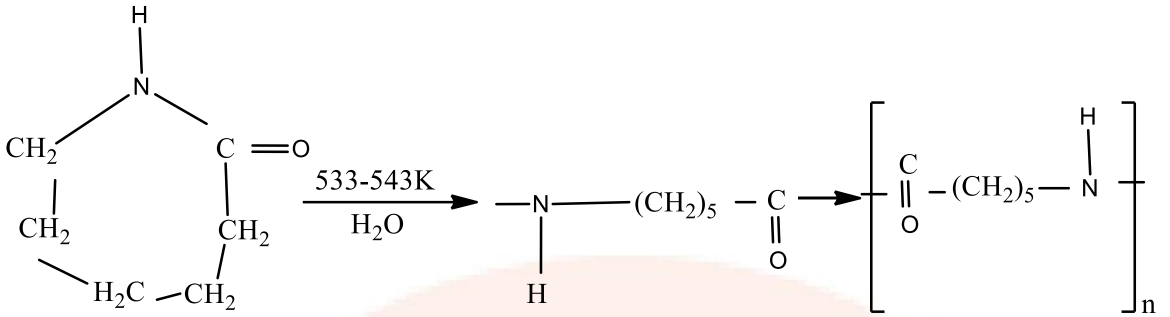
(b) 2- Nitropropane:

$$\underset{\underset{\text{(acetoxime)}}{\text{Iso-Propyldoxime}}}{\overset{\overset{CH_3}{|}}{CH_3CH}-N-OH} \xrightarrow{\text{Triflouroacetic acid}} CH_3-\overset{\overset{CH_3}{|}}{C}=\overset{\overset{O^-}{|}}{N}-OH \rightleftharpoons \underset{\text{2-nitropropane}}{CH3-\underset{\underset{H}{|}}{\overset{\overset{CH_3}{|}}{C}}-\overset{\overset{O^-}{|}}{N}=O}$$

(vii) The substance which are added in food to prevent its retardation or oxidative deterioration are called as antioxidants.



(viii) **Carbohydrates:**
Optically active polyhydroxy aldehyde or polyhydroxy ketones or the compounds that can be hydrolysed to polyhydroxy aldehyde or polyhydroxy ketones are called as carbohydrates.
Nylon-6:



Caprolactum Repeating Unit Nylon-6

Q.6. Answer any THREE of the following: [9]
(i) What are f-block elements? Distinguish between lanthanoids and actinoids.
(ii) Explain the terms
(a) Optical activity
(b) Ligand
(c) Interstitial compounds
(iii) Write the formula of Tetraamminedichloroplatinum(IV) chloride. How is propene converted into 1-bromopropane and 2-bromopropane?
(iv) What are broad-spectrum antibiotics? How are polythene and neoprene prepared?

Solution 6:
(i) **f-block elements:** The elements in which differentiating electron enters into (n-2)f orbital or the penultimate shell or antepenultimate shell of the d-orbital are called as f-block elements.

Difference between lanthanoides and actinoides

Lanthanoides	Actinoides
1. Differentiating electron enters in 4-f orbitals.	1. Differentiating electron enters in 5-f orbitals.
2. Belongs to 6^{th} period and forms a part of third transition series.	2. Belongs to 7^{th} period and forms a part of third transition series.
3. Binding energy of 4-f orbital is higher.	3. Binding energy of 4-f orbital is lower.
4. Only Promethium is radioactive.	4. All members are radioactive.
5. Beside +3 oxidation state lanthanoids shows +2 and +4 oxidation states in few cases.	5. Beside +3 oxidation state actinoids shows +4, +5, +7 higher oxidation states also.
6. Lanthanoids show low tendency to form complexes.	6. Actinoids show greater tendency to form complexes.
7. Some of the ions of lanthanoids are fairely coloured.	7. Most of the ions of actinoids are fairely coloured.
8. Lanthanoid hydroxides are less basic in nature.	8. Actinoid hydroxides are most basic in nature.
9. Contraction is relatively less.	9. Contraction is greater.

(ii) (a) **Optical activity:** The phenomenon or property of organic substance to rotate the plane of PPL towards right or left is called as optical activity.
(b) **Ligand:** The molecule or ion which are coordinated to the metal atom or ion in coordination compound are called as ligands.
(c) **Interstitial Compound:** Those compound which are formed when small amount of small atoms like H,C and N are trapped inside the interstitial space in crystal of metals.

(iii) The formula of Tetraamminedichloroplatinum(IV) chloride is
$[PtCl_2(NH_3)_4]Cl_2$
Propene converted to 1-bromopropane

$$CH_3 = CH_2 \xrightarrow{\text{Peroxide}} \underset{\text{1-bromopropane}}{CH_3CH_2CH_2Br}$$

Propene converted into 2- bromopropane

$$CH_3 = CH_2 \xrightarrow{\text{MKrule}} \underset{\text{2-bromopropane}}{CH_3CHBrCH_3}$$

(iv) Those antibiotics which are effective against a wide range of gram positive and gram negative bacteria are known as broad- spectrum antibiotics.
Polyethene:

$$nCH_2 = CH_2 \xrightarrow{\text{Polymerization}} \underset{\text{Reapeating unit}}{-CH_2 - CH_2 -} \rightarrow \underset{\text{Polyethene}}{\left(CH_2 - CH_2 \right)_n}$$

Neoprene:

$$nCH_2 = \underset{\underset{Cl}{|}}{C}CH = CH_2 \xrightarrow{\text{Polymerization}} \underset{\text{(Repeating Unit)}}{-CH_2 - \underset{\underset{Cl}{|}}{C}CH - CH_2 -} \rightarrow \underset{\text{Neoprene}}{\left(CH_2 - \underset{\underset{Cl}{|}}{C}CH - CH_2 \right)_n}$$

Q.7. Answer any ONE of the following:
(i) Explain the mechanism of esterification. Write the reactions involved in dehydration of 1°, 2° and 3° alcohols.
(ii) What are vitamins? Name any two diseases caused by deficiency of vitamin A. Write the structures of:
a. nucleoside b. nucleotide
How are 1-nitropropane, 2-nitropropane and 2-methyl-2-nitropropane are distinguished from each other using nitrous acid?

Solution 7:
(i) The esterification of carboxylic acid with alcohol is nucleophillic acyclic substitution. The carboxylic acid is protonated on its carboxyl oxygen atom. Alcohol acts as a nucleophile and attacks carboxyl carbon thus loss of proton gives ester hydrates.

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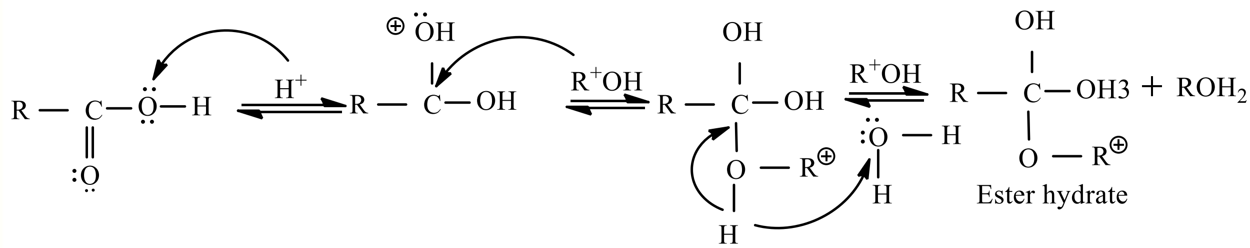
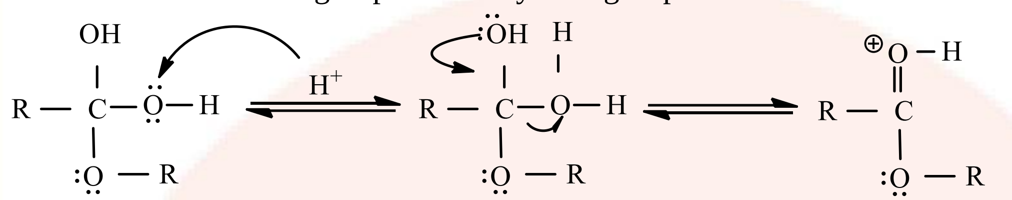
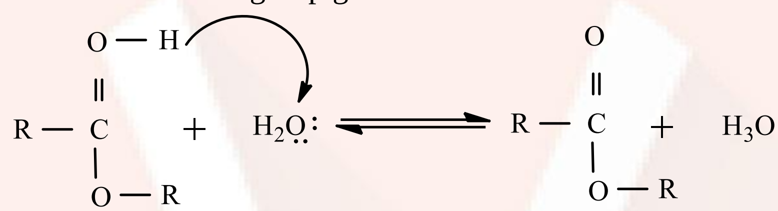


Further one of the –OH group of ester hydrate gets protonated and loses a molecule of water.



The loss of proton from second –OH group gives ester



Reactions involved in dehydration of 1°, 2° and 3° alcohols.

Primary alcohol is dehydrated by heating with 95% H_2SO_4 at 443K.

$$\underset{\text{Ethanol}}{H-\underset{H}{\overset{H}{\underset{|}{\overset{|}{C}}}}-\underset{H}{\overset{H}{\underset{|}{\overset{|}{C}}}}-O-H} \xrightarrow[443k]{95\% H_2SO_4} \underset{\text{Ethene}}{H-\overset{H}{\overset{|}{C}}=\overset{H}{\overset{|}{C}}-H}$$

Secondary alcohol is dehydrated by heating with 60% H_2SO_4 at 373K.

$$\underset{\text{Propane-2-ol}}{H-\underset{H}{\overset{H}{\underset{|}{\overset{|}{C}}}}-\underset{OH}{\overset{H}{\underset{|}{\overset{|}{C}}}}-\underset{H}{\overset{H}{\underset{|}{\overset{|}{C}}}}-H} \xrightarrow[373k]{60\% H_2SO_4} CH_3-CH=CH_2+H_2O$$

$$\underset{\text{Butane-2-ol}}{CH_3-\underset{OH}{\underset{|}{C}}-CH_2-CH_3} \xrightarrow[373k]{60\% H_2SO_4} \left[\begin{array}{l} \underset{\text{But-1-ene(Minor Product)}}{CH_3CH_2CH=CH_2} \\ \underset{\text{But-2-ene(Major Product)}}{CH_3CH=CHCH_3} \end{array}\right. + H_2O$$

Tertiary alcohol is dehydrated by heating with 20% H_2SO_4 at 373K.

$$\underset{\text{2-methyl-2-propanol}}{CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-OH} \xrightarrow[373k]{20\% H_2SO_4} \underset{\text{2-methyl prop-2-ene}}{CH_3-\underset{CH_3}{\overset{CH_2}{\underset{|}{\overset{\|}{C}}}}} + H_2O$$

(ii) Vitamins are the organic substance that must be supplied to permit proportionate growth in living being or for the maintenance of the structure.

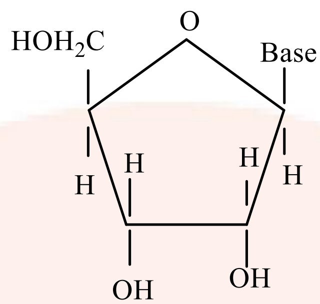
Diseases: 1. Night blindness.

2. Retardation of growth.

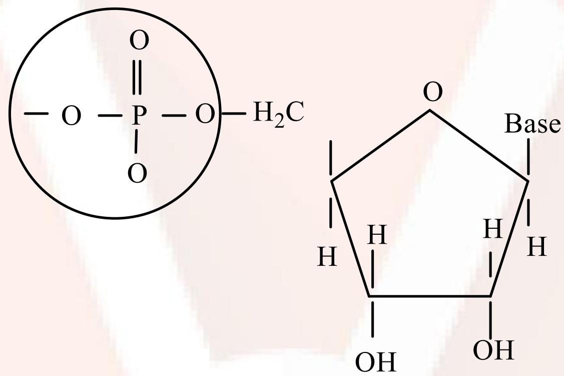
3. Dryness of hair and skin.

Structure:

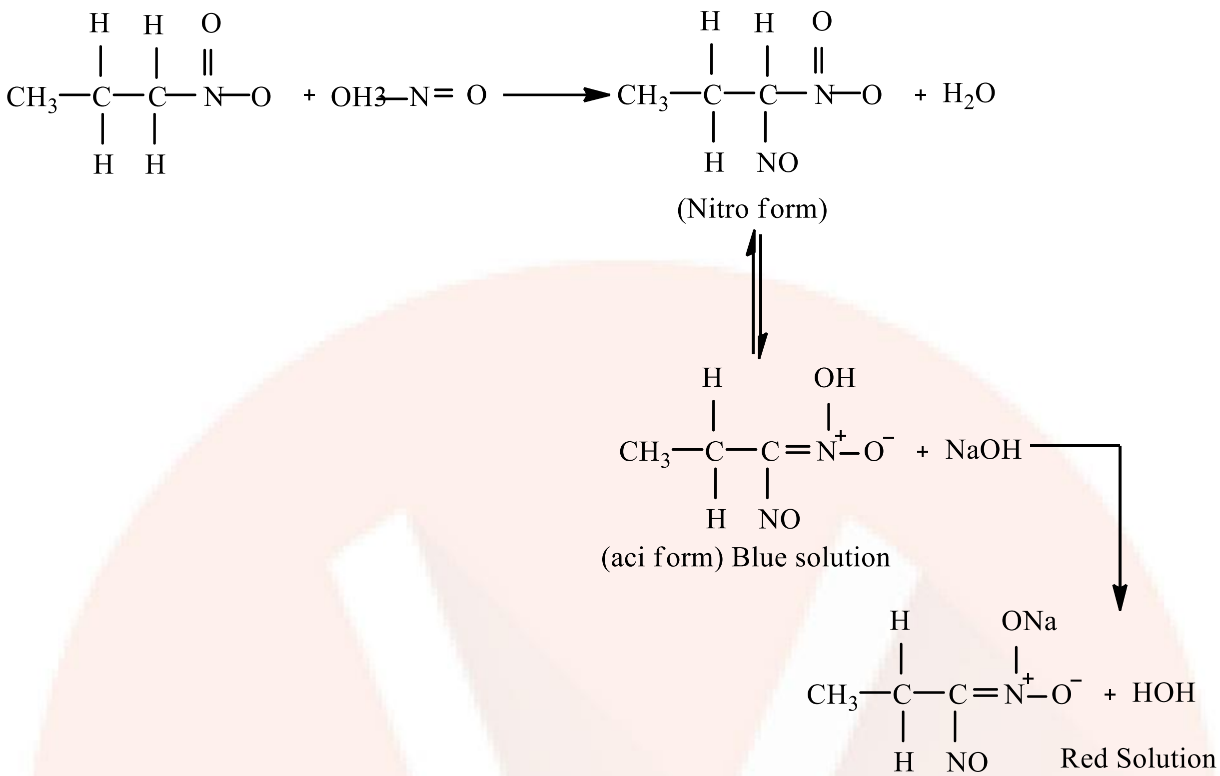
(a) Nucleoside



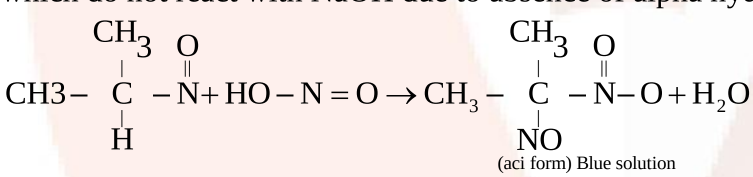
(b) Nucleotide



Primary nitroalkane reacts with nitrous acid to form blue coloured nitroso-nitropropane which reacts with NaOH to give red color solution.



When 2-nitropropane is treated with Nitrous acid it gives Blue coloured nitro-nitrosopropane which do not react with NaOH due to absence of alpha hydrogen.



2-methyl 2-nitropropane does not react with nitrous acid due to absence of alpha hydrogen.

Q8. Select and write the most appropriate answers from the given alternatives: [7]

(i) The preparation of alkyl fluoride from alkyl chloride, in presence of metallic fluorides is known as _______.

(A) Williamson's reaction (B) Finkelstein reaction
(C) Swarts reaction (D) Wurtz reaction

(ii) Identify the weakest acidic compound amongst the following:
(A) p-Nitrophenol (B) p-Chlorophenol (C) p-Cresol (D) p-Aminophenol

(iii) On acid hydrolysis, propanenitrile gives _______.

(A) propanal (B) acetic acid
(C) propionamide (D) propanoic acid

(iv) Which of the following amines yield foul smelling product with haloform and alcoholic KOH?
(A) Ethylamine (B) Diethylamine
(C) Triethylamine (D) Ethylmethylamine

(v) Which of the following is NOT present in DNA?
(A) Adenine (B) Guanine
(C) Thymine (D) Uracil

(vi) Amongst the following, identify a copolymer.
(A) Orlon (B) PVC
(C) PHBV (D) Teflon

(vii) Phenelzine is used as an _______.
(A) analgesic (B) antiseptic
(C) antipyretic (D) antidepressant

Solution 8:
(i) (c) Swarts reaction.
(ii) (d) p-Aminophenol
(iii) (d) Propanoic acid
(iv) (a) Ethylamine (BONUS)
(v) (d) Urasil
(vi) (c) PHBV
(vii) (d) Antidepressant.

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